

Social Media Platform Usage & User Motivations

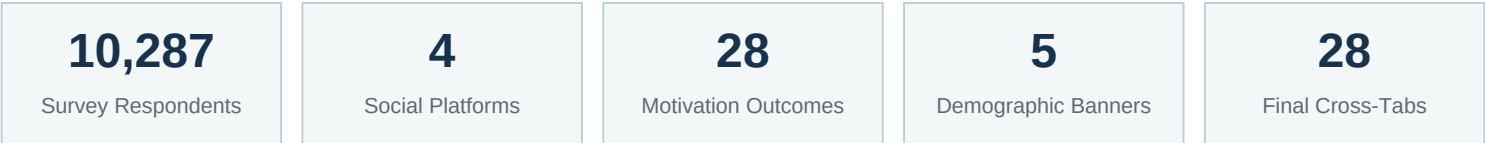
Survey Tabulation & Data Analytics Portfolio Case Study | Carlos Henrique Nakagomi

Independent portfolio analysis using Pew Research Center public-use data - not affiliated with or commissioned by Pew Research Center.

Research Objective

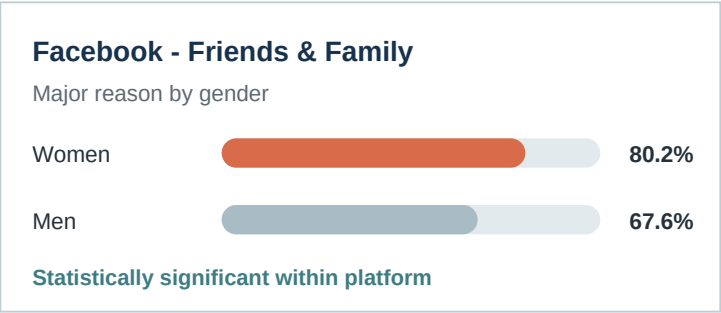
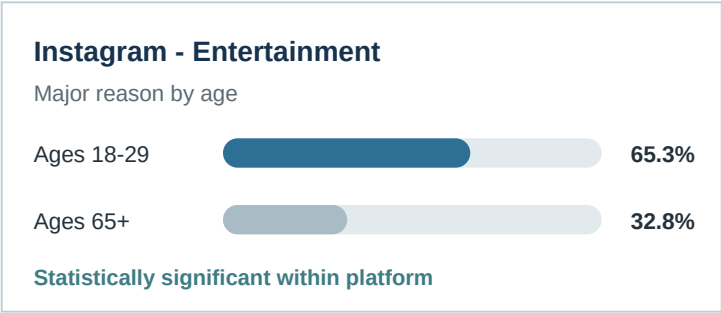
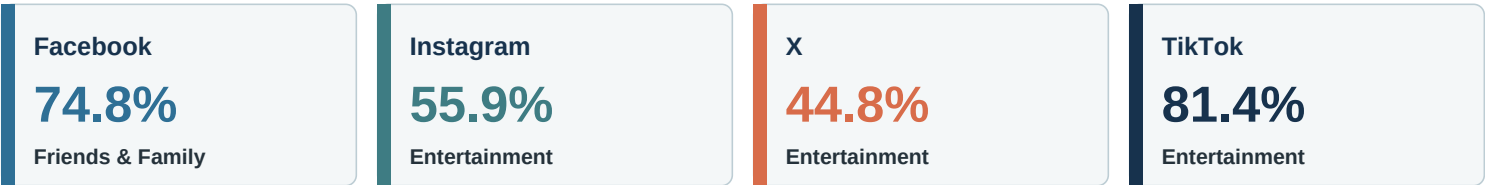
How do social media usage patterns and motivations differ across Facebook, Instagram, X, and TikTok, and across demographic groups?

Routed survey microdata became weighted cross-tabs, demographic comparisons, significance tests, QA, and an Excel tab book.



Key Findings

Weighted percentage identifying the motivation as a Major reason



X descriptive motivation cluster

Shared interests, sports/pop culture, news, and politics range from **24.3% to 28.3%** among X users. Entertainment leads at **44.8%**.

Interpretation guardrail

Platform percentages use different platform-specific or routed analytical universes. **Cross-platform comparisons are descriptive; significance testing is conducted within platform.**

Dataset: Pew Research Center American Trends Panel Wave 144, fielded March 18-24, 2024. Analytic file: 10,454 records, including 10,287 survey respondents.

Sources: final tables T05, T11, T15-T18, T20, T25; qa/significance_test_results.csv; <https://www.pewresearch.org/dataset/american-trends-panel-wave-144/>

Survey Tabulation, QA & Technical Delivery

Documented universes, weighted estimation, statistical testing, and independent output validation

Survey Tabulation Workflow

Questionnaire

Routing & Universe Validation

Variable Selection

Weighting

Cross-Tabs

Effective Bases

Significance Testing

QA

Excel Tab Book

Platform Universes & Weights

Platform	Analytical Universe	Survey Weight
Facebook	DOV_ASKFB_W144 == 1	WEIGHT_W144_FB
Instagram	DOV_ASKIG_W144 == 1	WEIGHT_W144_IG
X	SMUSE_c_W144 == 1	WEIGHT_W144_XT
TikTok	SMUSE_i_W144 == 1	WEIGHT_W144_TT

Facebook and Instagram use routed module universes. X and TikTok use their respective platform-user universes.

Statistical Testing

Measure	Project procedure
Banner design	Age, Gender, Education, Census Region, Income
Estimates	Weighted column percentages; unweighted bases
Variance input	Kish-adjusted effective sample sizes
Comparisons	Approx. two-sided weighted proportions
Testing floor	Minimum unweighted N = 100
Base flags	Small: N = 30-99; Very small: N < 30

Guardrail: No multiplicity correction is applied. Tests are exploratory and do not replicate Pew Research Center's complete complex-survey variance methodology.

Quality Assurance

504 / 504
Percentage-sum validations

1,512 / 1,512
Persisted weighted percentages reconciled

504
Effective-base cells validated

1,596
Pairwise significance tests

495
Significant comparisons

342
Unique significance-marker cells

Base-size review	476 reviewed	448 OK	21 small	7 very small	0 with N < 100 tested
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Final Tab Book

T11 — Instagram: Entertainment							
Response	Total	Age				Gender	
	Total	A. 18-29	B. 30-49	C. 50-64	D. 65+	A. A man	B. A woman
Unweighted Base	4496	542	1967	1217	764	1601	2841
Major reason	55.9%	65.3% BCD	58.2% CD	49.4% D	32.8%	56.7%	55.2%
Minor reason	31.0%	27.7%	30.5%	33.5%	36.3% AB	31.4%	31.2%
Not a reason	13.1%	7.0%	11.3% A	17.1% AB	30.9% ABC	11.9%	13.6%

Letters indicate significantly higher percentages than the referenced category within the same banner (two-sided 95% test). * N=30-99; ** N<30. Categories with N<100 are excluded from significance testing.

Representative enlarged crop of T11 - Instagram entertainment, showing Total, Age, and Gender banners.

Technical Delivery & Reproducibility

Technology	Role in the project
Python / pandas / NumPy	Validation, tabulation, weighting, significance testing, and QA
openpyxl / Excel	Automated final tab book and workbook round-trip validation
Git / GitHub	Version control and reproducible project delivery
Source-controlled evidence	Methodology, tabulation plan, QA outputs, and report builder

Final technical deliverable: output/tab_books/Reach3_Social_Media_TabBook.xlsx

Full project repository: <https://github.com/CarlosNakagomi/Social-Media-Survey-Tabulation>